

Splunk: Detecting Exfiltration (DNS)

Objective

The objective of this lab is to use Splunk to detect data exfiltration activities over DNS, ensuring timely identification and mitigation of unauthorized data transfers.

Steps

1. Detecting Exfiltration (DNS)

Used Splunk to monitor and analyze DNS traffic, focusing on identifying patterns and anomalies that indicate potential data exfiltration activities, such as unusual DNS queries or unexpected data patterns.



The screenshot shows the Splunk Search interface with a search query for DNS exfiltration. The query is: `index=dns_exf sourcetype="bro:dns:json" | eval len_query=len(query) | search len_query>=40 AND query!="*.ip6.arpa*" AND query!="*amazonaws.com*" AND query!="*_googlecast.*" AND query!="_ldap.*" | bin _time span=24h | stats count(query) as req_by_day by _time, id.orig_h, id.resp_h | where req_by_day>60 | table _time, id.orig_h, id.resp_h, req_by_day`. The search results show 17,116 events. The table below displays the results:

_time	id.orig_h	id.resp_h	req_by_day
2021-08-26 17:00:00	192.168.38.104	192.168.38.102	17116

2. Finding the Domain Associated with Data Exfiltration

Analyzed the detected DNS traffic to identify the specific domain associated with the data exfiltration activities.



The screenshot shows the Splunk Search interface with a search query for domains associated with data exfiltration. The query is: `index=dns_exf sourcetype="bro:dns:json" | eval len_query=len(query) | search len_query>=40 AND query!="*.ip6.arpa*" AND query!="*amazonaws.com*" AND query!="*_googlecast.*" AND query!="_ldap.*" | bin _time span=24h | stats count by query`. The search results show 17,116 events. The table below displays the results:

query	count
post.1001acea0.3d26fb65d7d.456c54f2.blue.letsghunt.online	4
post.106ab5e0b.3163696a29.456c54f2.blue.letsghunt.online	4
post.10f450a67.1c055a8630e.456c54f2.blue.letsghunt.online	4
post.1112690d8.19075866603.456c54f2.blue.letsghunt.online	4
post.1112695dc.11218e810d6.456c54f2.blue.letsghunt.online	4

Conclusion

Detecting data exfiltration over DNS is critical for protecting sensitive information from unauthorized access. By leveraging Splunk's analytical capabilities, organizations can detect and respond to exfiltration attempts, safeguarding their data and maintaining compliance with security policies.